College of Business
香港城市大學
City University of Hong Kong

FinTech and AI in Finance

Class 9: Stablecoins and CBDCs

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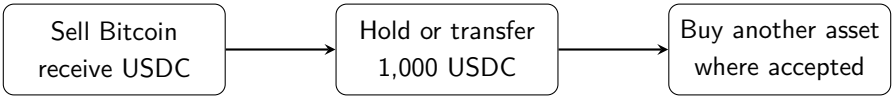
Semester A, 2026/27

City University of Hong Kong

Digital Money and Stablecoin Designs

Why Keep Dollars in USDC?

Hypothetical trader: sells Bitcoin for 1,000 USDC and wants to buy another asset later.



- The trader can stay in a dollar-denominated token between trades, without making a bank withdrawal each time.
- Moving to another platform requires a supported network. Getting bank dollars still needs a sale or eligible redemption.
- Circle targets US$1 per USDC; market prices can move away from that target.

Illustrative balance; ignore fees. Product reference: Circle, USDC, checked 28 October 2026.

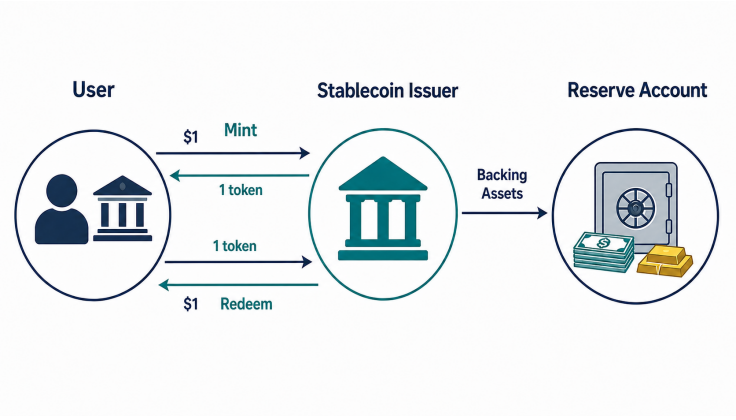
Bank Dollars and Wallet Dollars

A business wants to pay a supplier in dollars.

- **Bank deposit.** The business holds a claim on its bank and sends a payment through the banking system.
- **Reserve-backed stablecoin.** The business holds a token targeting $1 and can transfer it to a compatible wallet.
- **Returning to a bank account.** The holder needs an eligible redemption route or a buyer. Fees and processing time affect the amount received.

A dollar price target alone does not make a token a bank deposit.

Mint and Redemption Are Two Sides of One Promise



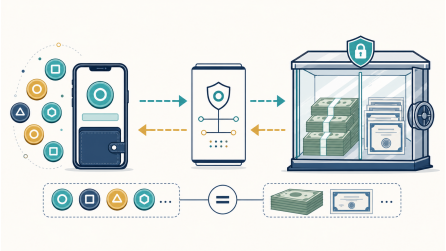
A reserve-backed stablecoin remains credible only if users can reverse the minting transaction and receive the promised cash value.

What Does a Stablecoin Promise?

The product aims to keep a token close to a reference value, usually one US dollar.

- **Price target.** Traders buy and sell the token at market prices, which can differ from $1.
- **Support for the target.** Reserves, crypto collateral, or trading incentives provide different forms of support.
- **Access to cash.** Redemption terms determine who can convert directly, at what cost, and how long processing takes.

Trace Four Functions End to End



1. ISSUE / REDEEM

Who creates tokens, and who converts them back to money?

2. HOLD BACKING

Where is it kept, valued, and segregated?

3. TRANSFER / SETTLE

Which ledger records ownership, and when is finality?

4. GOVERN / SUPERVISE

Who changes rules, freezes activity, or handles failure?

The practical test

The promise is only as strong as the weakest function in the chain.

Three Stablecoin Designs



OFF-CHAIN RESERVE

Cash and safe assets → reserve and redemption risk

Control: issuer and regulators

CRYPTO-COLLATERAL

On-chain collateral → price and liquidation risk

Control: contracts and governance

ALGORITHMIC

Incentives and demand → confidence and death-spiral risk

Control: contracts and governance

A CBDC belongs in a separate comparison because it is central-bank money, not a privately stabilized token.

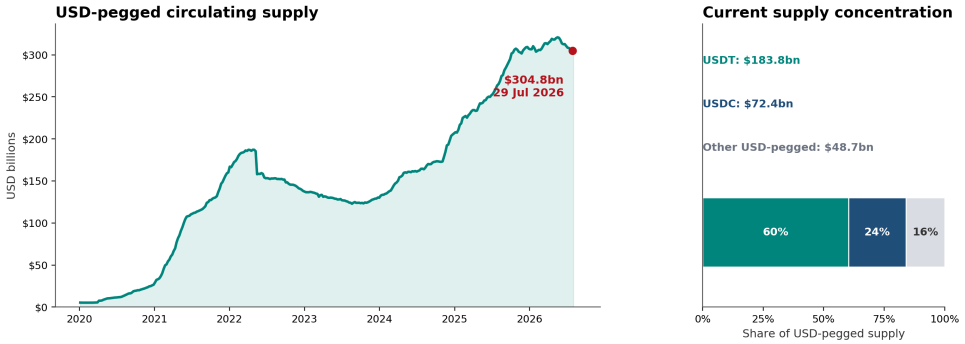
A Stablecoin Payment Ends When the Supplier Can Use the Money

Illustrative payment: a business already holds 100 USDC. Ignore fees and assume a $1 conversion price.

- **Send.** The business transfers 100 USDC to the supplier's compatible wallet.
- **Hold or spend.** The supplier can keep the tokens or use them for another accepted token payment.
- **Convert.** If wages must be paid from a bank account, the supplier needs a sale or eligible redemption that delivers bank money.

Confirmation of the token transfer and availability of bank funds are different steps.

Stablecoin Supply Grew and Remains Concentrated



Source: DefiLlama stablecoin APIs, through 29 July 2026. Values measure circulating USD-pegged supply, not transaction volume, reserves, or issuer equity.

Why the concentration matters

Stablecoins reduce the need to return to bank money between on-chain transactions, but users still depend on a small number of issuers, reserve portfolios, custodians, and redemption channels.

Allaire and Ardoino: Two Businesses behind Dollar Tokens



Jeremy Allaire

Circle co-founder and CEO

- Circle Mint serves eligible institutions.
- Banks and payment firms can connect bank dollars with USDC.
- Distribution partners expand access and share the income.



Paolo Ardoino

Tether CEO

- USDT is distributed through exchanges and wallets.
- Traders can hold dollar value between crypto trades.
- A small holder usually exits through a market sale.

Both businesses need liquid reserves and reliable routes back to bank money.

Sources: Circle Mint; Circle 2025 annual report; Tether product information. Checked 28 October 2026. Editorial cartoon portraits.

A Tokenized Fund Is Not a Stablecoin

Investment: ChinaAMC fund shares

The investor holds an interest in a portfolio of short-term assets.

Income after expenses accrues to investors. Fund value and secondary-market price can change.

Payment: a reserve-backed stablecoin

The holder seeks a transferable token that maintains its currency peg.

Redemption rights and access depend on issuer terms. Reserve income need not accrue to holders.

Connecting the two: stablecoins can pay for tokenized investments where supported. Asset ownership and the means of payment remain different claims.

Accepted settlement currencies depend on the product and platform.

Fund reference: ChinaAMC announcement, 23 October 2026.

Two Collateralized Models

- **Off-chain reserves (USDC, USDT):** an issuer holds cash or traditional assets and handles redemption.
- **Main risk:** reserve losses or delays in converting reserves into cash.
- **Crypto collateral (DAI, LUSD):** a protocol locks collateral worth more than the stablecoins issued.
- **Main risk:** falling collateral prices and delayed liquidation.

Algorithmic Stablecoins



- Supply adjustments and trading incentives aim to maintain a $1 peg.
- Designs such as TerraUSD/LUNA rely on another token rather than a full pool of fiat reserves.
- If confidence in both tokens falls, redemptions can accelerate the collapse. TerraUSD failed in 2022.

Stablecoin Issuers, Failures, and Policy

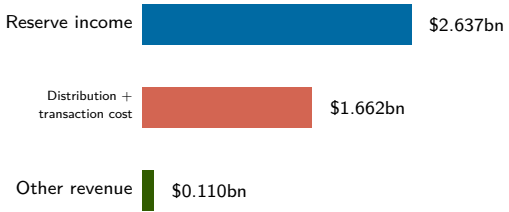
Terra: How a Reflexive Peg Failed

Promise	Mechanism	Hidden dependence
1 UST near $1	burn UST and mint $1 worth of LUNA	LUNA must retain market value and liquidity
Arbitrage restores peg	traders buy discounted UST for redemption	they must expect LUNA can still be sold
Protocol absorbs runs	more LUNA is issued as UST exits	supply expands exactly when confidence is falling

Why this old case stays

A claim cannot be stabilized by another claim whose value depends on confidence in the first one.

Circle: Reserve Income and Distribution Costs



- circulation and short rates drive reserve income;
- exchanges and other partners drive distribution but share the economics;
- payments and services can diversify revenue, but were still small in 2025.

Source: Circle Internet Group 2025 Form 10-K. Distribution plus transaction cost was about 63% of reserve income; this is not a profit margin.

Current Company Contrast: Tether and Circle

Business block	Tether / USDT	Circle / USDC
Distribution	broad global wallet, exchange, and on-chain liquidity	exchange, institutional, developer, and payment partnerships
Revenue engine	return on a very large reserve portfolio	reserve income plus smaller service and integration revenue
Reported 2025 scale	over $186bn issued; company reported over $10bn group profit	$2.637bn reserve income; public-company Form 10-K
Disclosure	quarterly reserve attestations; company announced KPMG's 2025 financial-statement audit in August 2026	audited financial statements plus reserve disclosures
Strategic risk	reserve mix, redemption, regulation, and confidence	interest rates, partner bargaining power, circulation, and regulation

The tokens may both target one dollar, but distribution, disclosure, and reserve-income sharing differ.

Sources: Tether 2025 Q4 figures and 13 August 2026 audit announcement; Circle 2025 Form 10-K. Tether audit status follows the company's announcement.

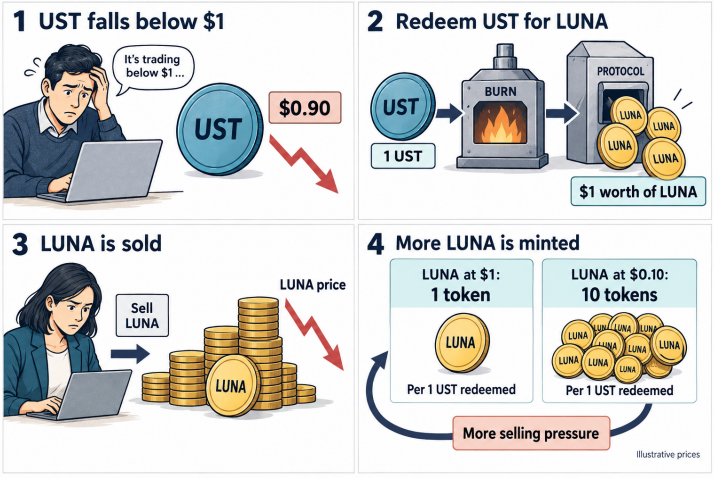
Circle: Dollar Payments outside Bank Hours



- Exchanges and payment firms need to move dollar value across markets and time zones. Bank cutoffs can delay the cash leg.
- Circle issues USDC against dollar reserves. Eligible businesses use Circle Mint to convert between bank dollars and tokens.
- Wallets can transfer USDC on supported networks around the clock. Receiving dollars in a bank account still requires redemption or a market sale.

Sources: Circle payments; USDC terms. Checked 13 October 2026.

UST and LUNA: The Redemption Spiral



Historical mechanism, May 2022. Source: SEC, Terraform complaint (2023), para. 33.

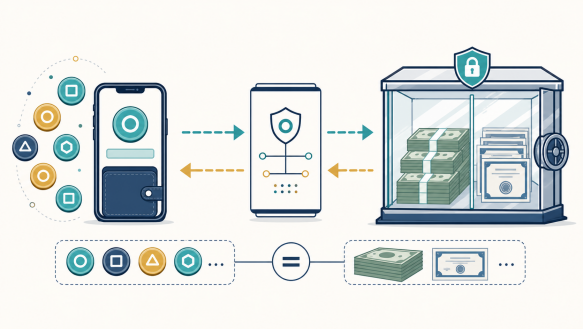
The Loss Spread through Several Balance Sheets

Peg breaks → forced redemptions → volatile token issuance → collateral losses → liquidations and withdrawals

- Stablecoin holders lost when the promised dollar value disappeared.
- Holders of the risk-absorbing token lost as supply expanded into falling demand.
- Protocols using either token as collateral inherited the price shock.
- Funds and lenders with concentrated exposures faced margin calls and funding stress.

Case lesson
Contagion follows shared collateral, leverage, and funding links. The blockchain label by itself carries none of it.

USDC: Redemption or a Market Sale?



- **Direct redemption:** eligible customers return USDC to Circle for dollars. Tokens leave circulation.
- **Market sale:** another buyer pays. Tokens remain in circulation, possibly below $1.
- Redemption depends on eligibility and bank processing. A sale depends on market depth.

Circle Mint route under applicable USDC terms. Eligibility and legal rights vary by jurisdiction. Fees may apply.

USDC and SVB: A Treasury Decision

During the March 2023 SVB crisis, uncertainty over access to Circle's bank deposits put USDC's redemption promise under pressure.

Hypothetical CFO: holds 1m USDC and must pay $800,000 on Monday. A dealer offers an executable price of $0.90 per USDC, with dollars available by the payment deadline.

- How many USDC must the company sell to fund the payment?
- What must be true for waiting for redemption to be feasible?
- Could a short-term credit line be cheaper than selling at a discount?

Company, quote, and payment terms are teaching assumptions, not historical market data. Ignore fees. Historical sources appear on the next page.

USDC and SVB: What Changed on 12 March 2023?

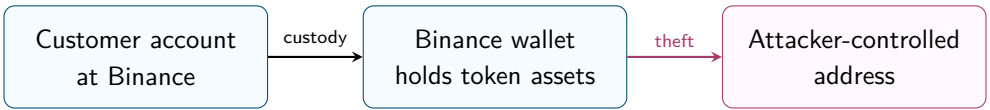
- Circle had **$3.3bn** of USDC reserves deposited at SVB, about 8% of total reserves.
- On **12 March**, US authorities announced protection of all SVB depositors, with access to funds from **13 March**.
- Circle said its SVB reserves would be fully available. Access to bank money changed the outlook for USDC redemption.

The action protected SVB depositors. It did not create government deposit insurance for USDC holders.

Sources: Circle statement dated 12 March 2023; Treasury, Federal Reserve, and FDIC joint statement; USDC terms, no deposit insurance.

Binance: Reserve Backing and Custody Risk

30 October 2026, HKSAR time: Binance reported about **US$6.3bn** stolen from hot wallets, which are connected to the internet.

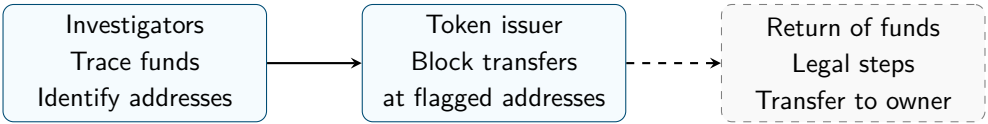


- Circle and Tether reserves back their tokens; they do not protect an exchange account against theft.
- Binance paused withdrawals and pledged to cover the loss with SAFU and company funds.

Simplified custody relationship. Source: Binance, 30 October update, checked 1 November 2026.

Who Can Freeze Stolen Stablecoins?

Elliptic reported that almost all assets frozen or seized after the Binance theft were **USDT and USDC**.



- BTC and native ETH have no issuer with an equivalent token-freezing power. Swapping into them can make recovery harder.
- A freeze can limit losses, but it also shows the issuer's control over transfers. **Frozen does not mean returned.**

Simplified recovery process. Sources: Elliptic, Binance investigation; Binance, 30 October update. Checked 1 November 2026.

Multiple-Choice Question: Reserve Income after Distribution Costs

Suppose a stablecoin issuer holds \$1 billion of reserves earning 3% annually. It pays distribution partners 60% of that reserve income.

How much annual reserve income remains before the issuer's other expenses?

A) \$18 million

B) \$30 million

C) \$12 million

D) \$60 million

Multiple-Choice Question: The Full Cost of a Stablecoin Payment

A US company pays an overseas supplier. A stablecoin route costs $1 for transfer, $8 for currency conversion, and $4 for redemption. A bank quotes an all-in cost of $10 for the same amount received.

Which route is cheaper, and by how much?

A) The bank route, by $3.

B) The stablecoin route, by $9.

C) The stablecoin route, by $1.

D) Both routes cost $10.

Stablecoin Reserves and Treasury Demand

The GENIUS Act, signed on 18 July 2025, establishes a framework for payment stablecoin issuers and their reserves.

What the law requires

- at least one dollar of eligible liquid reserves per dollar issued;
- monthly reserve reports, examined by a registered accounting firm.

Where that money goes

- Tether reported about US$141bn of direct and indirect Treasury-bill exposure at 31 March 2026;
- Circle holds most USDC reserves in a government money-market fund.

Sources: GENIUS Act (P.L. 119-27), 18 July 2025; issuer disclosures.

Worked Example: Reserve Income at Two Interest Rates

Illustrative annual figures: Reserves stay at US$100 million. Distribution and operating costs total US$2 million. Holders receive no interest.

Scenario A: 4% reserve yield

$$100\text{m} \times 4\% = \text{US\$4m income}$$

$$4\text{m} - 2\text{m} = \boxed{\text{US\$2m profit}}$$

Scenario B: 2% reserve yield

$$100\text{m} \times 2\% = \text{US\$2m income}$$

$$2\text{m} - 2\text{m} = \boxed{\text{US\$0 profit}}$$

- These are operating profits before tax, assuming reserve interest is the only revenue.
- Token circulation is unchanged, but the lower interest rate removes the issuer's profit in this example.

Multiple-Choice Question: Redemption Arbitrage

A stablecoin trades at \$0.98. An eligible trader buys 1,000 coins and immediately redeems them for \$1 each. Total trading and redemption fees are \$5.

What is the trader's net profit?

A) \$0

B) \$20

C) \$25

D) \$15

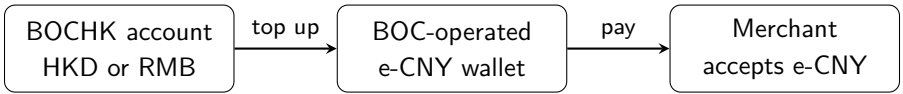
Policy Questions for China

Policy objective	Stablecoin channel	Evidence needed
Capital-flow monitoring	wallets may move claims outside conventional bank channels	identity, location, transaction, and redemption path
Consumer protection	users may confuse a token with insured bank money	issuer, reserve, right to redeem, and loss allocation
Monetary sovereignty	foreign-currency tokens may affect local money demand	sustained balances and use, not wallet downloads alone
Innovation	token rails may reduce some payment frictions	end-to-end cost, compliance, liquidity, and recovery

The relevant boundary depends on the user, transaction, backing, and redemption route.

BOCHK and BOC: Funding an e-CNY Wallet

Customer need: an HKSAR resident wants to pay a merchant in the Chinese Mainland.



- BoC Pay+ links the customer to a wallet operated by Bank of China. BOCHK provides the exchange and top-up service.
- Funding from HKD requires currency conversion; wallet limits and merchant acceptance still apply.
- Easier payment supports RMB use for this purchase. It does not establish wider demand to save or borrow in RMB.

Sources: BOCHK service terms; HKMA e-CNY guide. Checked 28 October 2026.

HKSAR Stablecoins: Two Distribution Models

First two issuer licences: 10 April 2026

Anchorpoint Financial

Partner distribution

Authorized distributors bring HKDAP to institutional and professional investors.

At 30 August: limited rollout on Ethereum, with access through distributors.

HSBC Hong Kong

Existing bank customers

Plans to use PayMe and the HSBC HK App for transfers and payments.

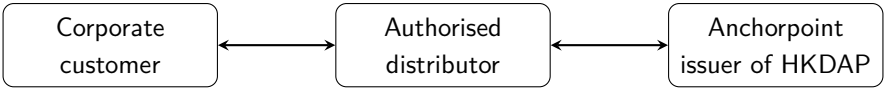
At 30 August: licensed, with launch still planned for the second half of 2026.

Discussion: Which route can attract repeat users at a lower distribution cost?

Sources: HKMA, 10 April 2026; issuer disclosures. Business status retains the course's 30 August 2026 snapshot.

Anchorpoint: Who Does What in an HKDAP Purchase?

Beta-access example: an eligible corporate customer wants HKDAP.



- **Distributor:** opens the customer account, checks identity, and provides the conversion interface.
- **Anchorpoint:** issues HKDAP, maintains backing, and processes the distributor's redemption request.
- **Custody:** the customer's chosen wallet or custodian controls access to the tokens. This is separate from holding the issuer's reserves.

Simplified access and redemption route. Sources: Anchorpoint FAQ and whitepaper, checked 28 October 2026.

Building Public and Regional Alternatives

Alternative	What it changes	What still determines adoption
Domestic CBDC	provides a digital form of public money	distribution, privacy, resilience, and user need
Multi-CBDC experiment	coordinates participating central banks and institutions	governance, FX liquidity, legal rules, and counterparties
Linked payment networks	connects existing domestic rails	fees, identifiers, compliance, exceptions, and reach
Tokenized deposits	represents commercial-bank money on programmable infrastructure	bank credit, interoperability, and redemption

Public infrastructure can reduce a specific friction; macro credibility and market depth remain separate.

CBDCs and the e-CNY

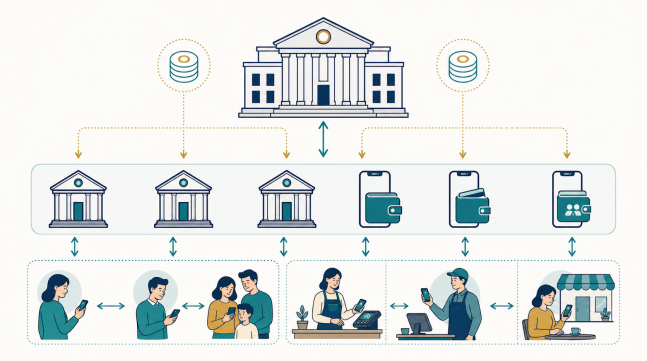
What is a CBDC?

- A **central bank digital currency (CBDC)** is a digital liability of the central bank.
- Retail CBDCs serve households and businesses. Wholesale CBDCs serve financial institutions.
- Unlike a private stablecoin, the holder's claim is on the central bank.

Why Central Banks Explore CBDCs

- **Domestic efficiency:** modernize payments and reduce some cash-handling costs.
- **Financial inclusion:** broaden access if wallets, identity, devices, and offline options are well designed.
- **Monetary sovereignty:** preserve access to public money as private digital payment instruments grow.
- **Cross-border settlement:** test whether shared infrastructure can reduce time, cost, and reconciliation.

CBDC Design Models



- the central bank issues the public money;
- banks and payment firms distribute wallets and serve customers;
- households and firms make payments without holding an account directly at the central bank.

The two-tier model preserves private-sector distribution.

CBDC Work Is Widespread, but Wholesale Projects Lead

93
central banks in the BIS 2025 survey

91%
exploring retail or wholesale CBDC in 2024

Track	Main unresolved design question
Wholesale	interbank and cross-border settlement, liquidity, and governance
Retail	distribution, privacy, offline access, bank funding, and adoption
mBridge	reached MVP stage in 2024 with Chinese Mainland and HKSAR participation, among others

Sources: BIS Papers No. 159 (August 2025); BIS, mBridge MVP announcement (June 2024).

If Deposits Move into a Retail CBDC, What Happens to the Bank?

Hypothetical bank: customers convert \$10m of deposits into a direct central-bank liability. Assume settlement uses the bank's central-bank reserves.

Customer balance sheet	**Commercial-bank balance sheet**
Bank deposits: $-\$10$m	Reserve assets: $-\$10$m
Retail CBDC: $+\$10$m	Deposit liabilities: $-\$10$m

- Equity is unchanged at conversion: this is a funding outflow, not an immediate \$10m loss.
- If outflows persist, the bank may need replacement funding or fewer assets. Holding limits can reduce the outflow.

Simplified retail-CBDC design, not the 2026 e-CNY bank-wallet arrangement discussed later.

Multiple-Choice Question: Is 100% Backing Enough?

A stablecoin issuer holds reserves equal to 100% of tokens, entirely in ten-year government bonds, while promising instant redemption at par. Which conclusion is best?

A) Full backing removes run risk.

B) Duration and liquidity losses can still impair redemption.

C) The market price must remain exactly at par.

D) Token holders directly own the bonds.

Discuss
Separate the reserve's face value from its market value and liquidity under mass redemption.

How Stablecoins May Undermine Monetary Sovereignty



- **Access:** global digital dollars can improve payment and saving options.
- **Leakage:** flows may move around domestic banks and capital controls.
- **Policy:** local money demand, seigniorage, and monetary transmission may weaken.

Convenience for users can become a sovereignty problem for the state.

Case 2: e-CNY — From Cash-Like Pilot to "2.0"

Phase	Positioning	Evaluation question
2014 research start	central-bank digital-currency design	which public-money problem should be solved?
Pilot era	cash-like public money through a two-tier architecture	can wallets deliver reach, privacy, resilience, and acceptance?
From 1 Jan. 2026	"2.0" treatment links bank-wallet balances more closely to deposit money	how do interest, reserves, insurance, and bank balance sheets interact?
Next stage	integration with existing interfaces and cross-border experiments	does the system create sustained use and system-level value?

Source: PBoC action-plan reporting via Xinhua / PRC Ministry of Justice, December 2025.

e-CNY Scale Is Large, but the Denominator Matters

Official cumulative measure	Reported scale	What it does not reveal
Transactions	3.32 billion	active monthly users or repeat-use distribution
Value	RMB 14.2 trillion	average balances or share of total payments
Personal wallets	225 million	active, funded, or primary wallets
Pilot coverage	26 localities in 17 provincial-level regions	merchant depth or displacement of Alipay and WeChat Pay

Use cases include retail payment, government disbursement, offline or hardware wallets, and cross-border experiments. Source: State Council of the PRC, official e-CNY statistics, 29 October 2025.

Short-Answer Question: Who Bears the Wallet Outage?

A retail CBDC is accessed through a commercial-bank wallet. The app fails for a day, but the central bank remains solvent.

1. Identify the issuer of the monetary claim and the operator of customer access.
2. Explain which risk remains even though the central bank is solvent.
3. Name one responsibility or control that should be assigned before an outage occurs.

Answer in three or four sentences
Separate claim safety from operational availability and recovery.

Why Would a WeChat Pay User Try an e-CNY Wallet?

Hypothetical shopper: already uses WeChat Pay and Alipay for daily purchases.

- **At the checkout.** If the existing app works and the merchant accepts it, another wallet must offer a useful reason to switch.
- **A one-off reward.** A sign-up subsidy may attract a first payment. What brings the shopper back after it ends?
- **A different need.** The BOCHK/BOC example helps an eligible cross-border customer fund an e-CNY wallet. Its value should be judged for that customer.

Compare repeat payments, total cost, and failed transactions for the same shopping task.

Shopper and reward are teaching assumptions. Cross-border service: BOCHK terms.

Short-Answer Question: Wallet Registrations and Payment Use

In a retail digital-money pilot, registered wallets double after a one-off signup reward. Monthly active users remain unchanged, and merchants report little repeat use.

1. Does the registration increase establish wider use for payments? Explain using the evidence given.

2. Choose two measures of sustained payment use. How would you use them to assess adoption after the reward ends?

The 2.0 Framework Changes the Evaluation Questions

Evaluation area	Question	Evidence
Bank funding	does interest-bearing wallet money compete with or strengthen deposits?	balances, flows, funding cost
Safety net	how do reserves, insurance, wallet tiers, and responsibility apply?	rule text, failure and recovery tests
Interoperability	can one public instrument support several customer interfaces?	acceptance, switching, and uptime
Cross-border use	does the pilot reduce friction while respecting capital, FX, identity, and data rules?	end-to-end time, cost, exceptions

The next test

Judge it on sustained use, balances, and system-level value. Wallet count alone says very little.

Tether: Dollar Access through Crypto Markets



- Traders need a dollar-denominated asset they can move between exchanges. International users may also face costly or limited access to dollar bank transfers.
- Tether issues USDT and relies on exchanges, wallets, and market makers for distribution. Most small users obtain tokens through these intermediaries.
- The service connects dollar value with crypto trading and payments. A user's exit price depends on both market liquidity and confidence in redemption.

Source: Tether, how it works; direct-access conditions. Checked 13 October 2026.

Tether: Who Can Redeem Directly?



- Tether's published minimum for direct redemption is **$100,000**, with a fee of the greater of **$1,000 or 0.1%**. Verification is required.
- A small holder typically sells to another market participant. That sale transfers existing USDT rather than removing it from circulation.
- If dealers cannot redeem quickly or cheaply, market prices may deviate from $1. Reserve liquidity and the redemption channel both matter.

Source: Tether fee schedule, checked 13 October 2026.

Tether: Earnings and Reserve Risk



- Tether reported **$1.5bn net operating profit in Q2 2026** and a **$4.11bn reserve buffer** at 30 June 2026.
- Reserve investments can generate income while tokens circulate. Lower interest rates can reduce that income, even without losing users.
- The buffer and asset liquidity protect different risks. A reserve snapshot does not show how quickly every asset can be sold during heavy redemptions.

Source: Tether, Q2 2026 financial figures. Company-reported figures, not comparable to Circle's gross reserve income.

What to Remember

1. Reserve-backed stablecoins need liquid assets to meet redemptions. For example, **Circle** uses cash and short-dated Treasuries.
2. Reserve income and distribution costs are central to issuer profitability. For example, **Circle pays distribution costs to Coinbase**.
3. CBDCs are central-bank liabilities, with retail or wholesale access. For example, **mBridge** uses wholesale CBDCs for interbank settlement.

Sources: Circle, reserve disclosures; Circle, 2025 annual report; HKMA, 2024 annual report.